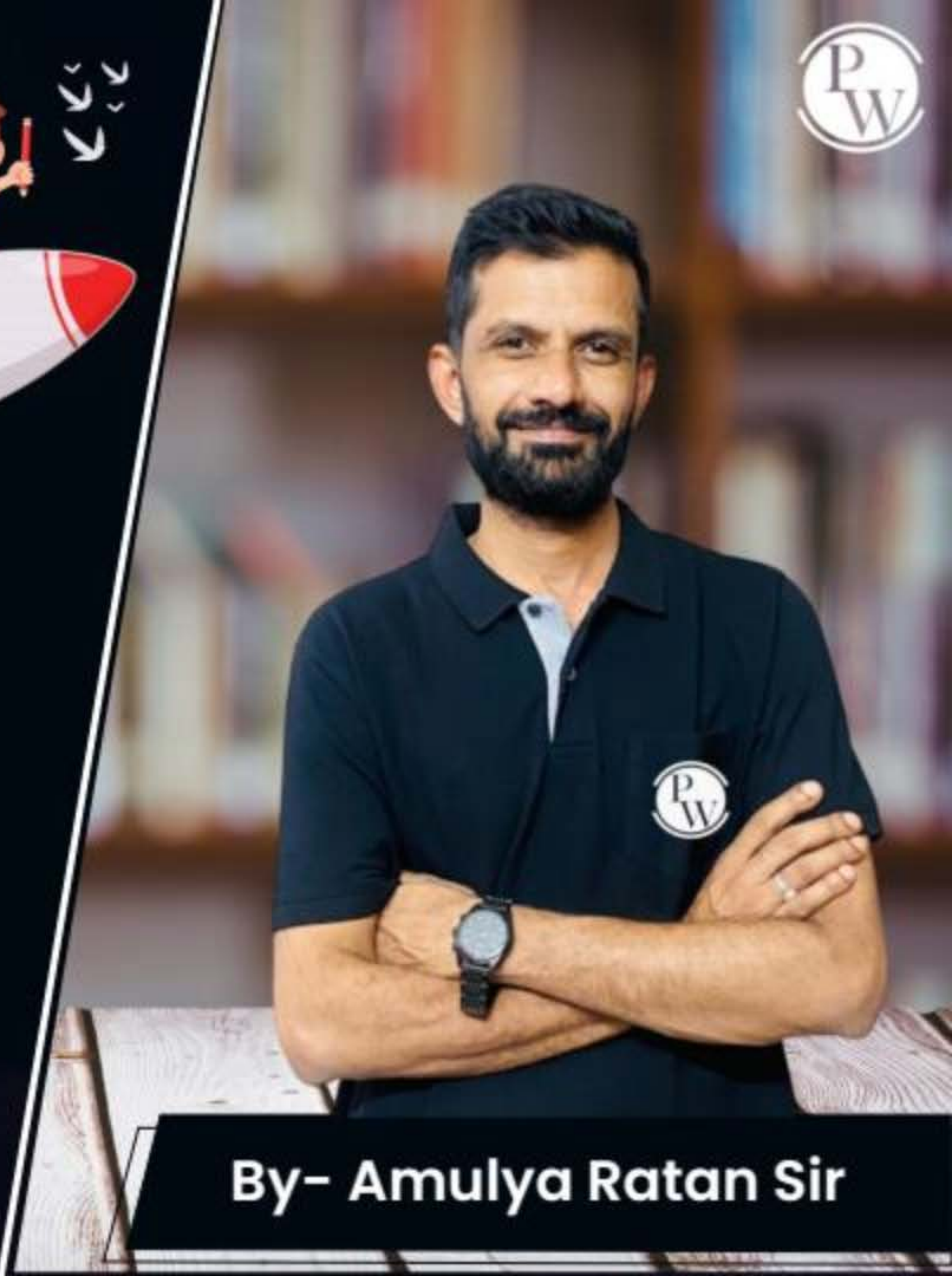
GATE

ALL BRANCHES

General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 01



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Topics to be Covered

Topic-1 Calendar (English)

CALENDAR

(Solar)

+1

+17

Use ?

Basic Questions

Monday ?(1) 01-01-01

Date → Day ?

1. Use of CALENDAR?

(4 months) Only

2. How many months in a year consist of 30 days?

11 months

April
June
Sep
Nov

3. Which is the first day of a week?

Sunday

0

4. What is the difference between A.D. & B.C.?

00-00-00

Anno Domini (Holy Year)

1600 B.C. (Before Christ)

LEAP YEAR Occurrence

Q. Maximum ⇒ 8 years

29th Feb

1646 ✗ (N.Y)

1442 ✗ (N.Y.)

1211 ✗ (N.Y.)

1642 C.E. (A.D.)

(Common Era)

B.C.E. (B.C)

(Before Common Era)

1744 (L.Y.)

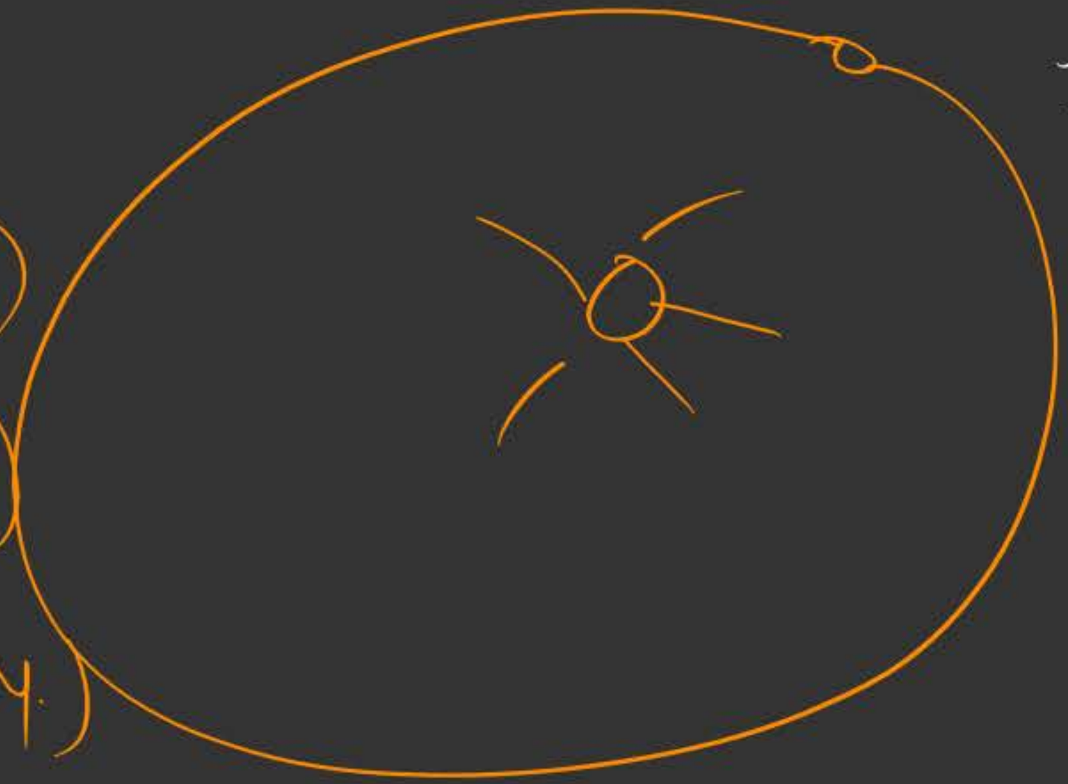
Divide by 4

1 → 365 days (N.Y)

2 → 365 days (N.Y)

3 → 365 days (N.Y)

4 → 365 + 1 = 366 days (L.Y)



400?

~~$365\frac{1}{4}$ days~~

~~OR~~

~~365 days 6 hrs?~~

365 days 5 hrs 32 min ... seconds

6 hrs

6 hrs

6 hrs

6 hrs

24 hrs

Century

400

29th Feb, 1796

1800 (Not) (N.Y)

29th Feb, 1804

8 years

To be Noted:

✓If a year is divisible by 4 then it's a leap year else normal. e.g. 1652, 1212, 1496, 1708, etc. are leap years whereas 1714, 1446, 2006 etc. are normal years.

✓If a century is divisible by 400 then it's a leap year else normal. e.g. 1600, 1200, 2000, 800 etc. are leap years whereas 1000, 1500, 2100 etc. are normal years.

13 04
08
12
16
20
24
28
1

ODD DAYS

?

$7\overline{)80}(11$
77
$3 \Rightarrow$ odd days

Q If As today is 6^{th} May, 2025. The day is Tuesday. +3

✓What would be the day on 16^{th} May, 2025? ?

FRIDAY

$7\overline{)50}(7$
49
$1 \Rightarrow$ odd days

10 days

3 days

7

ODD DAYS

$\boxed{0 - 6}$

✓ All those number of days which can't be kept in a group of a week.

OR

✓ When given number of days is divided by 7, the remainder is called as odd days.

Normal Year & Leap Year

52 weeks

100 years → ?

N.Y. ⇒ 365 ⇒ 1 odd day

L.Y. ⇒ 366 ⇒ 2 odd days

First Century

100 years → 5 odd days

L.Y. / N.Y.

24 / 76

×100 ⇒ 5 odd days | 500

×200 ⇒ (10) 3 odd days | 600

×300 ⇒ (15) 1 odd day | 700

✓ 400 ⇒ ~~(20) ⇒~~ 6 + 1 = 7

0 odd days | 800

×2 → 48

×1 → 76

6 + 6 = 12

900 - 5
1000 - 3
1100 - 1
1200 - 0

04
08
12
16
20
24
28
32

1

96

×24

5008

Odd Days in Centuries

100 - 5

+5

200 - 3

300 - 1

400 - 0

+5

500 - 5

500	900	1300	1700
600	1000	1400	1800
700	1100	1500	1900
800	1200	1600	2000

Questionnaire:

Thursday

5th Aug, 2004

2003 → 3 odd day

2000 → 0 odd day

3 N.Y → 3 odd days

3+1 → 4 odd days

(31) J → $\cancel{3}$

(29) F → $\cancel{1}$

M → $\cancel{3}$

A → $\cancel{2}$

M – $\cancel{3}$

J – $\cancel{2}$

J – 3

(5) A – 5

→ 1 odd day

0 → Sunday

1 → Monday

2 → Tue

3 → Wed

4 → Thurs

5 → Fri

6 → Sat

Questionnaire:

2 odd days

3rd June, $\boxed{1947}$ ⇒ Tuesday

04

1946 ⇒ 2 odd days

1900 → 1 odd day

46 → 1 odd day ×11

L.Y N.Y

11 35

22 35

1 + 0 = 1

44

J - ~~3~~ A - ~~2~~

F - 0 M - ~~3~~

M - ~~3~~ J ⇒ ~~3~~

0

Questionnaire:

#Q. Guru Nanak was born on 15th April, 1469. What was the week day?

4 odd days

Thursday

1468 ⇒ 4 odd days

1400 ⇒ 3 odd days

68 ⇒ 1

L.Y 17, N.Y 51

34, 51

6 + 2 = 8

$\times 17$

04, 68

J → 3
F → 0
M → 3
A → 1

0

Fact:

April 15, 1469: Day of the Week

April 15, 1469 was the 105th day of the year 1469 in the Gregorian calendar. There were 260 days remaining until the end of the year. The day of the week was *Thursday*.

The day of the week for April 15, 1469 under the old Julian calendar was Saturday. Did you notice the difference with the Gregorian calendar?

Questionnaire:

$\boxed{2001 \Rightarrow 1}$

$\boxed{13^{th} \text{ Dec}, 2002}$ FRIDAY

J - $\cancel{3}$ J - $\cancel{2}$

F - 0 J - $\cancel{3}$

M - $\cancel{3}$ A - $\cancel{3}$ D ⇒ $\cancel{6}$

A - $\cancel{2}$ S - $\cancel{2}$

M - $\cancel{3}$ O - $\cancel{3}$

N - $\cancel{2}$

4

$2000 \rightarrow 0$

$1 \rightarrow 1$

$1 + 4 = 5$

[MCQ]

#Q. Mahatma Gandhi was born on 2nd October 1869. What was the week Day?

Assignment

A Monday

B Thursday

C Tuesday

D Saturday

[MCQ]

#Q. If January 1st, 1992 was a Wednesday. What day of the week was January 1st, 2003?

Assignment

A Sunday

B Wednesday

C Thursday

D Friday

#Q. A person was born on fifth Monday of February in a particular year. Which one of the following statements is correct based on the above information?.

Assignment

A The 2^{nd} February of that year is a Tuesday.

B There will be five Sundays in the month of February in that year.

C The 1^{st} February of that year is a Sunday.

D All Mondays of February in that year have even dates.

2 mins Summary

Topic Calendar

Date $\rightarrow$ Day ?

THANK - YOU